

**Department of Statistics and Business Analytics
STAT 482 - Capstone in Analytics for Business**

Spring 2026

Case Study 7: Building an Interactive Business Analytics Dashboard Using ChatGPT and Excel Data

Due submission date: Wednesday, April 29, 2026

Brief Description

A mid-sized retail company wants to better understand customer behavior and operational performance using data stored in Excel. The dataset includes both **quantitative variables** (e.g., sales, delivery time, customer ratings) and **qualitative variables** (e.g., region, product category, customer segment).

Student teams act as analytics consultants. They must use **ChatGPT as a support tool** and develop a solution (preferably in **R Shiny**) that:

- Reads an Excel dataset
- Handles mixed data types
- Produces **on-demand descriptive statistics and summaries**
- Presents results through an interactive dashboard

Students must critically evaluate ChatGPT outputs and ensure correctness.

Objectives

By completing this case, students will:

1. Apply data cleaning and preprocessing techniques to real-world Excel data
2. Distinguish and appropriately handle **quantitative vs qualitative variables**
3. Use ChatGPT to assist in:
 - Designing workflows
 - Writing/debugging code
 - Structuring dashboards
4. Build an **interactive dashboard** (using R Shiny or equivalent)
5. Generate:
 - Summary statistics (mean, median, SD, etc.)
 - Frequency tables for categorical variables
6. Translate analytical outputs into **business insights**
7. Demonstrate awareness of **data ethics, quality, and limitations**

Recommended Solution Methodology

Step 1: Problem Framing

- Identify business questions (e.g., “Which region has the highest sales variability?”)
- Define dashboard users and decisions supported

Step 2: Data Preparation

- Import Excel data (readxl in R)
- Clean:
 - Missing values
 - Inconsistent categories
- Classify variables:
 - Numeric
 - Categorical
 - Ordinal / text

Step 3: ChatGPT-Assisted Design

Students use ChatGPT to:

- Suggest dashboard layout
- Recommend useful statistics
- Generate starter R code

Step 4: Dashboard Development (R Shiny)

Core features:

- File upload (Excel)
- Variable selection
- Automatic detection of variable type
- Outputs:
 - Descriptive statistics (numeric)
 - Frequency tables (categorical)
 - Basic plots (histograms, bar charts)

Step 5: Validation

- Cross-check statistics manually or in Excel
- Test edge cases (missing data, small samples)

Step 6: Interpretation

- Convert results into business insights
- Provide actionable recommendations

Expected Deliverables

1. Case Report (Written)

Includes:

- Problem definition
- Data description and cleaning steps
- Dashboard design explanation
- Statistical outputs and interpretation
- Reflection on ChatGPT use
- Limitations and ethics

2. Oral Presentation

- Demonstration of dashboard
- Key insights
- Explanation of methodology
- Q&A discussion

Optional R Tools Students Can Use

- readxl (Excel input)
- dplyr (data manipulation)
- ggplot2 (visualization)
- shiny (dashboard)
- DT (interactive tables)